



## MEASUREMENTS, SIGNIFICANT FIGURES, AND ERRORS

### 1.1 Program Outcomes (POs) Addressed by the Activity

At the end of the term, the students must be able to

- Ability to apply knowledge of computing fundamentals, computing specialization, and mathematics, science, and domain knowledge appropriate for the computing models.
- Ability to function effectively as a leader and as a member or leader in diverse teams and in multidisciplinary settings.
- Ability to communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

### 1.2 Activity's Intended Learning Outcomes (AILOs)

At the end of this activity, the student should be able to

- define measurement, significant figures, and sources of error,
- conduct accurate measurements,
- apply the rules in determining significant figures,
- analyze errors.

### 1.3 Objectives of the Activity

At the end of this activity, the student should be able to

- to measure length using ruler, vernier caliper and micrometer,
- to determine the limit of accuracy of the different measuring devices,
- to differentiate the precision of measurements and accuracy of measurements, and
- to apply some elementary quantitative tools to evaluate measured quantities.

### 1.4 Principle of the Activity

The history of systems of measurement can be traced to the first recorded civilizations in the world until the systems were made and established to be more cohesive in the 18<sup>th</sup> century. Ancient people would use their foot, hand, arm to measure lengths of objects around until the eventual inventions of devices such as rulers, calipers, and others through the centuries.



To formally establish the universal system of measurements, in 1960 the *Système International d'Unités* or International System of Units (**SI**), commonly known as the metric system, was established by the 11<sup>th</sup> General Conference of Weights and Measures. The "SI unit" for length is meter; gram for mass; liter for volume/capacity; and Newton for force, to mention a few. The SI uses the prefixes for powers of 10 for length, volume, and mass: a kilogram means 1000 grams; a milliliter (mL) means  $10^{-3}$  of a liter; and a centimeter for  $10^{-2}$  of a meter.

On the other hand, the Imperial or the English system uses foot (length), inch (length), pound (mass and force), and gallon (capacity). These two systems are widely used depending on the nature of the objects being measured.

Alongside technological advancement, scientists have established the concepts of accuracy and precision across lab experiments to ensure any significant deviation would be avoided to keep the data exact or at least approaching the true value. A certain device could measure 100 cm but another could render 100.04 cm, but the true value is 101 cm. Such a scenario in the lab is inevitable but can be prevented with the correct use of appropriate tools, and further knowledge on basic measurement.

Some tools used in measuring small lengths or distances are vernier calipers and micrometers.

### ***Micrometer Caliper***

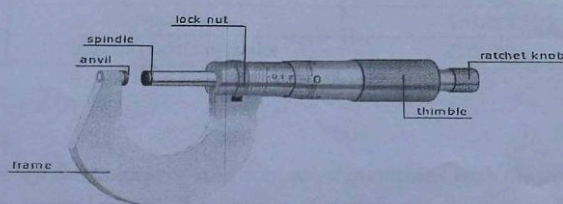


Figure 1.1. Parts of a Micrometer.

The micrometer is used to measure solid objects accurately that may fit in its spindle and anvil. It is calibrated using the knob. It gives a reading in millimeters. *Inventor: William Gascoine (1612-1644), England.*

### ***Vernier Caliper***

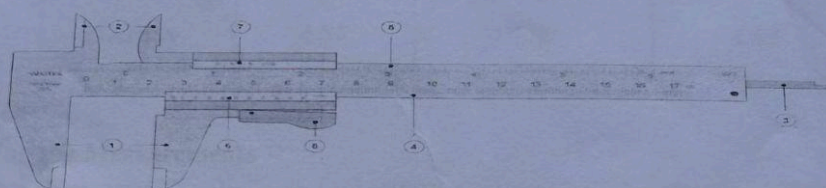


Figure 1.2. Parts of a Vernier Caliper.



**Legend:**

1. Lower Jaw - measures outer dimensions
2. Upper Jaw - measures inner dimensions
3. Blade - measures depth
4. Main Scale (in centimeter unit) - the fixed scale
5. Main Scale (in inch unit) - the fixed scale
6. Vernier Scale (centimeter unit) - the sliding scale or the movable scale
7. Vernier Scale (inch unit) - the sliding scale or the movable scale
8. Knurled Wheel - sliding jaw

The vernier caliper is best used for measuring hollow right circular cylinders as it can measure outer/inner dimensions, and depth. *Inventor: Pierre Vernier (1584-1638), France.*

**1.5 Materials/Equipment**

| Qty | Item                    | Group          | Class |
|-----|-------------------------|----------------|-------|
| 1   | ruler                   | Measuring tool | 0     |
| 1   | Vernier caliper         | Measuring tool | 0     |
| 1   | Micrometer caliper      | Measuring tool | 0     |
| 1   | beam balance            | Measuring tool | 0     |
| 1   | marble                  | Others         | 0     |
| 1   | hollow plastic cylinder | Others         | 0     |

**1.6 Procedure/s**

*Remark: Record the measured and computed quantities with the correct number of significant digits. For measured values, the number of significant digits is as follows:*

| Quantity | Ruler | Vernier Caliper | Micrometer |
|----------|-------|-----------------|------------|
| Diameter | 3 SF  | 4 SF            | 4 SF       |
| Height   | 3 SF  | 4 SF            | n/a        |
| Density  | 4 SF  | 4 SF            | 4 SF       |

**1.6.1 Getting the Measurements**

1. Using a ruler, vernier caliper, and micrometer caliper, measure the dimensions of the plastic cylinder and marble, respectively. Perform this step three times. Record measured values in Table 1.7.1.



2. Calculate the mean (or simply average) using the equation below:

$$\text{Mean} = \frac{\text{sum of measurements}}{\text{number of measurements}}$$

3. Record in Table 1.7.1.

### 1.6.2 Deriving Quantities (Volume and Density)

1. Using the beam balance, determine the mass of the **hollow cylinder**. Record this value to Table 1.7.2.
2. From Table 1.7.1, copy the mean diameter ( $\bar{D}$ ) values and mean height ( $\bar{H}$ ) values for this cylinder to Table 1.7.2.

3. Compute the volume of the hollow cylinder using the equation below:

*Note: Use the  $\pi$  on the calculator. Do not use 3.14 or 3.1416 as it would affect the accuracy of the calculation.*

$$V_{\text{cylinder}} = \frac{\pi (\bar{D}_{\text{out}}^2 - \bar{D}_{\text{in}}^2)}{4} \cdot \bar{H}$$

4. Using the formula that defines density, compute for the experimental density.

$$\rho_{\text{exp}} = \frac{\text{mass of the object}}{\text{volume of the object}} = \frac{m}{V}$$

5. Record all values to Table 1.7.2.
6. Determine the mass of the marble using the beam balance. Compute the volume of the marble and record to Table 1.7.2.

$$V_{\text{marble}} = V_{\text{sphere}} = \frac{\pi}{6} \cdot \bar{D}^3$$

7. Compute the density using the above formula, and the percentage error. Record all values to Table 1.7.2.
8. Compare this experimental density to the theoretical density provided in Table 1.7.2. Compute for the percentage error using the following formula:

$$\% \text{ Error} = \left| \frac{\rho_{\text{theo}} - \rho_{\text{exp}}}{\rho_{\text{theo}}} \right| \times 100$$